Meiyi Dong

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post-graduate

EDUCATION

City University of Hong Kong QS Top100 — Aug 2023 - Oct 2024

Multimedia Information Technology - Department of Electronic Engineering — Hong Kong, China

GPA: 3.78

Classification: Distinction

professional knowledge involved:

Machine Learning: MLE, MAP, BDR, LS, GMM-EM, SVM, LR, Naive Bayes, Bayesian networks, Neural Networks, PCA, KDE and other algorithms including mathematical derivation and code implementation and project application, Pytorch, Sklearn and other tools.

Data Mining: decision tree construction and evaluation, clustering algorithm (K-means,K-means++ ,Density-based clustering,Hierarchical clustering), anomaly detection (model-based,clustering-based,density-based), association rule mining.

Security Technology: encryption algorithm (DES,AES,RSA, etc.), network security (IPSec, etc.), authentication protocol design

Data Engineering: crawler, data preprocessing, data visualization, database (SQL), Transformer structure and application, recommend algorithm

Complex Networks: network models (small-world, scale-free, etc.), graph theory, game theory

Information Security: number theory, key management, e-commerce system design, security standards

Hebei University of Technology 211 Double 1st - Class — Sep 2018 - Jun 2022

Computer Science and Technology - Department of Computer Science — Tian Jin, China

GPA: 3.59

Major Courses:

C Programming (98); Python Programming (90); Object-oriented Programming (94); Artificial Intelligence

Basics (95); Data Structure (90); Operating System (86); Calculus (95); Software Engineering (86),etc.

hornor: Second Class Scholarship awarded by Hebei University of Technology in 2019

RESEARCH

Towards Accurate and Interpretive Disease Diagnosis for Aging Adults Through Deep Learning Framework Dec 2023 - Aug 2024

City University of Hong Kong

Existing studies have shown that the human gut microbiome can be closely related to specific diseases. This research took this phenomenon as the starting point and focused on three common diseases of the elderly: Type 2 Diabetes, Alzheimer's Disease, and Sarcopenia to find out more about how microbiome would impact human health. By leveraging the classification ability of the machine learning models, this research aimed to build a solid model for disease diagnosis and explore a deeper connection between these three diseases and the human gut microbiome.

Accurate classifiers were built into the research. The microbiome abundance in different sample sets was taken as the training/testing data. The Encoder layers of the Transformer were used to learn hidden knowledge from the data, and neural networks were used as the classifier.

The deeper connection between gut microbiome and diseases was explored in the research. The application of SHAP and other methods to explore the relationship between microorganisms and specific diseases even further.

#This research was conducted under the guidance of Prof. Sun Yanni & Dr. Shang Jiayu in CityU

paper/data/model in this link: https://github.com/JustaBirdy/disease_diagnosis

PROJECT EXPERIENCE

Minitype Search Engine Oct 2023 - Dec 2023

The goal of the project is to make a search engine with targeted goals, sufficient data stock, and efficient retrieval. I completed the entire project cycle independently as a developer. From the application of Python script (beautifulsoup library, request library) to crawl the data of the source website StackOverflow.Com, Spacy for data preprocessing, FastText for word embedding of the data, BM25 algorithm to give weight to the text vector, cosine similarity formula to calculate the similarity between the search word and the text vector, and finally to the application of approximate proximity search algorithm annoy to construct a binary tree structure to optimize the search speed, finally, an average retrieval speed of 0.016 seconds in one million pieces of data was obtained.

Anomaly Detection in Stock Market Data Mar 2024 - Apr 2024

This project aims to use anomaly detection technology in data mining to discover and analyze abnormal data of certain company stocks in the stock market. First, the Yahoo Finance API was used to crawl the stock data of Apple, Microsoft, and Google for the past year, and then the daily change trend of each stock was calculated. The k-nearest neighbor method was used to calculate the anomaly score of each change and arrange them (the advantage of this method is that it does not need to assume that the data conforms to a certain distribution). Then, the date with the highest anomaly score can be obtained. Combined with the information published by the company and the data on the Internet, it can be checked to see the important events that occurred on these dates that affected the company's stock price.

Rumor Detection Model Based on TF-IDF Oct 2023 - Nov 2023

This project aims to build an English rumor detection model by using NLP technology to learn the rumor characteristics of social platforms. More than 2,000 short sentences with labels were collected from the well-known social platform Twitter. The specific labels include "fact", "rumor" and "unverified". The token library was built through data cleaning, data integration, word segmentation and other methods, and the word vector matrix was built using tfidfvectorizer. Finally, the word vector was input into the SVM algorithm to train a multi-class classification model to achieve text classification.

Clustering-based image segmentation Apr 2024 - May 2024

This project aims to segment images into a custom number of images through clustering methods. The three main algorithms used are: K-Means algorithm, GMM-EM algorithm and Mean-Shift algorithm. First, feature extraction is performed on each pixel of the image to construct the feature matrix of the image. Then, these three algorithms can be used to perform cluster analysis on the feature matrix and classify each pixel into a certain label. K-Means and EM algorithms are custom number of categories, while Mean-Shift is an adaptive number of categories (depending on the parameters of the kernel function).

Intelligent cloud manufacturing platform Aug 2022 - Jul 2023

This project mainly studies the industrial intelligent manufacturing of aircraft parts. I am involved in the development of the intelligent cloud manufacturing platform on the WEB side. The platform is mainly oriented to the interaction between the main engine factory and the supplier, and the interaction between the main engine factory and the supplier. The functions of the platform include but are not limited to the management of production plan, the quality management of manufactured workpieces, and the personnel management of the main engine factory and the supplier factory, the standard library and database of aircraft airworthiness, dynamic visualization of data in the workshop, etc.

In the work, HTML,CSS are used to build web page structure and style, JS is used to write web page logic, element-ui, echarts and other third-party component libraries are used to assist in development, and GITLAB code warehouse based on GIT is used for collaborative development with back-end personnel.

Work Experience

AVIC Airborne Common Technology Co., Ltd Aug 2022 - Jul 2023

Web Development Engineer Yang Zhou, China

I was engaged in Web development at AVIC Airborne Common Technology Co., Ltd. The company is a subsidiary of the Aviation Industry Corporation of China and has undertaken many major scientific research tasks. During my tenure, I participated in the web front-end development task of industrial intelligent manufacturing of airborne parts for scientific research projects and was responsible for cooperating with back-end developers to develop the data operation platform based on the Web. The specific functions included planning management, quality management, airworthiness management, airworthiness knowledge base, personnel management, and other functional modules. The front-end development of hundreds of web pages. The main technologies used in the work are HTML/CSS/JS, VUE framework, third-party component libraries such as Echarts data visualization and element-ui, and GitLab code management. I have plenty of experience in self-made web page materials and Windows desktop program development using Photoshop software and experience using SQL to build database interfaces.

I was highly praised by the leaders of the group and received the monthly bonus for many times due to excellent working performance and a decent attitude.

Extracurricular Activities

Young Volunteers Association in HeBei University of Technology Branch Oct 2018 - Sep 2020

Deputy Director of Organization Department

During my stay in the volunteer association, I acted both as an organizer and a participant. Together with other member volunteers, I put up a series of meaningful activities. I organized a fundraising on the street to collect donations for children in poverty-stricken areas in western China. In three months, I achieved a total of about 1,000 yuan. I was recognized and rewarded by the association, received thanks messages from children I helped, and won the title and honorary certificate of "excellent volunteer".

This experience improved my communication and organizational ability and most importantly, the volunteering activities helped those who needed help.

Martial Art Club of Hebei University of Technology Oct 2018 - Jun 2020

Trainee

I participated in the lectures teaching about martial arts and exercises. Through this experience, I got to know more about the chinese traditional culture and maintained my health. At the same time, I was also responsible for part of the planning work. With the other members in the club, we successfully held the 2019 Hebei University of Technology Traditional Martial Art Performance Competitio, which was praised a lot from the other students.

SKILLS

- **Programming:** Python, HTML/CSS, JavaScript, Java, SQL, Linux Shell, C, C++, C#
- **Certifications:** IELTS English (7.5) CET-6 (498) CET-4 (546)
- **Languages:** Mandarin (native), English (proficient)
- **Software:** Visual Studio, Visual Studio Code, Office, Adobe PS, Adobe PR
- **Interests:** Music, Board Games, Card Games, Badminton

SUMMARY

As a student, I have achieved excellent academic results in school. I ranked top 4% in the HeBei Province for the National College Entrance Examination. I have achieved outstanding GPA and have no record of failing courses throughout my undergraduate and postgraduate period. I won the university scholarship once. I scored 7.5 on the IELTS test and full marks in the reading section on my first try at the examination, having no trouble studying and working in an English-speaking environment. I have abundant project experience and have been the team leader in many of them.

As an employee, I have a strong sense of responsibility for any tasks assigned to me. With the cautious attitude and reasonable planning, I could always finish the job on time with high quality. I am also a quick learner for studying and applying new things whenever I encounter them at work. I have received praise from superiors and colleagues many times for my high action ability.

With all the education I received and the jobs I did, I have built a solid professional foundation for further education and research. I am fully prepared to meet the challenges in the future work.